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Phosphoproteomics of CD2 signaling reveals AMPK-dependent regulation of lytic granule polarization in cytotoxic T cells

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Understanding the costimulatory signaling that enhances the activity of cytotoxic T cells (CTLs) could identify potential targets for immunotherapy. Here, we report that CD2 costimulation plays a critical role in target cell killing by freshly isolated human CD8⁺ T cells, which represent a challenging but valuable model to gain insight into CTL biology. We found that CD2 stimulation critically enhanced signaling by the T cell receptor in the formation of functional immune synapses by promoting the polarization of lytic granules toward the microtubule-organizing center (MTOC). To gain insight into the underlying mechanism, we explored the CD2 signaling network by phosphoproteomics, which revealed 616 CD2-regulated phosphorylation events in 373 proteins implicated in the regulation of vesicular trafficking, cytoskeletal organization, autophagy, and metabolism. Signaling by the master metabolic regulator AMP-activated protein kinase (AMPK) was a critical node in the CD2 network, which promoted granule polarization toward the MTOC in CD8⁺ T cells. Granule trafficking was driven by active AMPK enriched on adjacent lysosomes, revealing previously uncharacterized signaling cross-talk between vesicular compartments in CD8⁺ T cells. Our results thus establish CD2 signaling as key for mediating cytotoxic killing and granule polarization in freshly isolated CD8⁺ T cells and strengthen the rationale to choose CD2 and AMPK as therapeutic targets to enhance CTL activity.

INTRODUCTION

The immune synapse, which is formed between a cytotoxic CD8⁺ T cell (CTL) and a target cell, is characterized by substantial intracellular reorganization. To kill their targets, CTLs must polarize both the microtubule-organizing center (MTOC) and lysosome-like organelles filled with perforin and granzymes, called “lytic granules,” toward the contact site. Lytic granules are transported along microtubules through a dynein-dependent mechanism toward the MTOC, which becomes docked beneath the synapse, to focally release their contents inside the synaptic cleft (1). Notwithstanding the importance of granule polarization in the killing process and immune surveillance, unexpectedly, little is known about the costimulatory and signaling pathways that regulate granule positioning at the MTOC. Those factors that have been implicated include the strength of T cell receptor (TCR) stimulation (2), engagement of the lymphocyte

function-associated antigen 1 (LFA-1) (3), the kinetics of intracellular Ca²⁺ flux (4), and CD103-dependent activation of phospholipase Cγ1 (PLC-γ1) (5). However, our overall knowledge remains fragmentary, and identifying the principal regulators of this process represents an important goal, which has important implications for the design of CTL-targeting immunotherapies.

One of the essential tools to gain insight into the biology of CTLs is to study freshly isolated CD8⁺CD57⁺ T cells, which are present in substantial numbers in the peripheral blood of healthy human donors (6, 7). The CD8⁺CD57⁺ effector cell population is highly enriched for perforin-positive CTLs with cytotoxic activity (7, 8); therefore, CTL function can be evaluated in vitro using either enriched CTLs or the bulk CD8⁺ T cell population without previous stimulation or expansion. Note that the phenotype and function of freshly isolated CTLs appear to reproduce the host phenotype more reliably than do CTLs expanded in vitro, as exemplified by studies on CTLs from patients and mice with genetic mutations that affect CTL function (9). In this context, we previously reported that freshly isolated human CTLs, but not in vitro-expanded CTLs, form dysfunctional immune synapses with human resting B cells characterized by defective granule polarization (6). Our study showed that the functioning of freshly isolated CTLs could be finely regulated during immune synapse formation by the uncoupling of granule movement from MTOC translocation toward the synapse, which leads to a dispersion of lytic granules to the CTL periphery with a concurrent loss of cytotoxic killing.

These observations provided us with a model with which to study the signaling pathways controlling the formation of functional lytic immune synapses between B cells and CTLs and those regulating granule polarization. Here, we addressed both issues by combining two quantitative proteomics workflows. First, we resolved the complexity

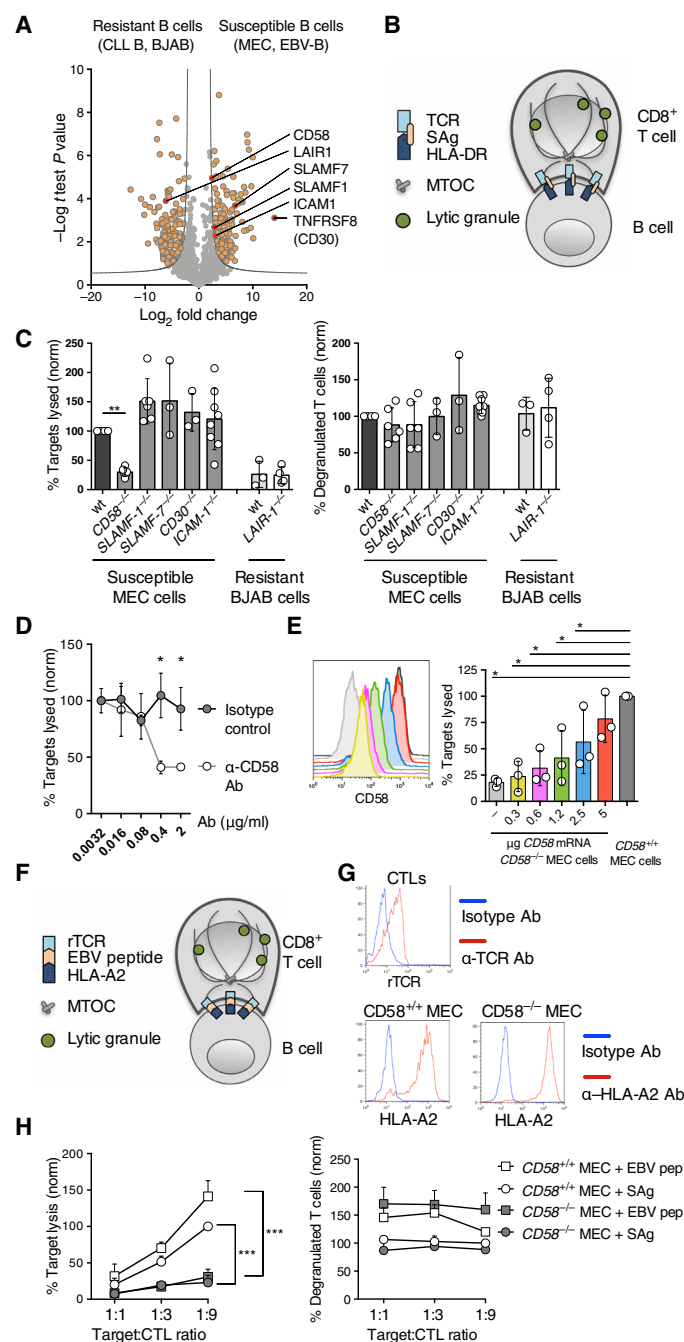
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Fig. 1. Proteomics analysis reveals CD58 as the determinant of B cell susceptibility to lysis by freshly isolated CD8⁺ T cells. (A) Surface proteome analysis of B cells, which were either resistant or susceptible to lysis. Resistant cells were B cells from patients with CLL, as well as a Burkitt lymphoma–derived cell line, BJAB (6). Susceptible cells were the CLL-derived B cell line MEC and an Epstein-Barr virus–immortalized B cell line (EBV-B). The volcano plot shows the log₂ fold change (resistant/susceptible B cells) in protein abundance plotted against the $-\log_{10} P$ value. Proteins with statistically significantly different abundances are highlighted either in orange or red [candidates screened in (C)]. (B) Experimental model for the assessment of CD8⁺ T cell cytotoxicity against SAg-loaded B cells. (C) Flow cytometric analysis of cytotoxicity (left) and degranulation (right) of CD8⁺ T cells incubated with the indicated CRISPR-Cas9–modified MEC and BJAB B cells. Data are means $\pm$ SD of three to six experiments with two mutant clones for *CD58*, *SLAMF-1*, and *ICAM-1* and one mutant clone for *SLAMF-7*, *CD30*, and *LAIR-1*. Values for wild-type (wt) MEC cells were set at 100% (killing ranged from 10.6 to 26.3% of targets; degranulation occurred in 3.8 to 8.8% of cells). (D) Flow cytometry analysis of CD8⁺ T cell killing of MEC cells in the presence of either isotype control antibody (Ab) or CD58-specific blocking antibody. Data are means $\pm$ SD of three experiments. Values for the treatment with isotype control antibody (0.0032 μ g/ml) were set as 100% (killing ranged from 23.2 to 41.8% of targets; degranulation occurred in 9.6 to 12.1% of cells). (E) Left: Flow cytometry analysis of *CD58*^{+/+} and *CD58*^{-/-} MEC cells, which were left untreated or were electroporated with increasing amounts of *CD58*-encoding mRNA. Right: Flow cytometry analysis of the cytotoxicity of CD8⁺ T cells incubated with the indicated target cells. Data are means $\pm$ SD of three experiments. Values for the *CD58*^{+/+} MEC cells were set as 100% (killing ranged from 19.7 to 33.3% of targets). For the experiments in (C) to (E), CD8⁺ T cells were incubated with target cells at a 5:1 ratio. (F) Experimental model for the assessment of killing by freshly isolated CD8⁺ T cells against peptide-loaded B cells. rTCR, recombinant TCR specific for EBV peptide. (G) Flow cytometry analysis of freshly isolated CD8⁺ T cells electroporated with mRNAs encoding recombinant *TCR α* and *TCR β* (beta) incubated with *CD58*^{+/+} and *CD58*^{-/-} MEC cells electroporated with *HLA-A2*-encoding mRNA. (H) Flow cytometry analysis of the cytotoxicity and degranulation of CD8⁺ T cells that were electroporated with mRNA encoding an rTCR specific for EBV peptide and incubated at the indicated ratios with MEC target cells, which were loaded with either an SAg mixture or EBV peptide. Data are means $\pm$ SD of four experiments. Values for the 1:9 ratio of SAg-loaded *CD58*^{+/+} MEC cells to CD8⁺ T cells were set as 100%. Killing ranged from 6.4 to 28.6% of targets, whereas degranulation occurred in 3.1 to 11.0% of cells. Data were analyzed by Mann-Whitney test. **P* < 0.05, ***P* < 0.01, and ****P* < 0.001.



of the surface proteome of B cells, establishing that CD8⁺ T cell costimulation through the surface receptor CD2 interacting with CD58 on B cells was critically important for granule polarization toward the immune synapse and efficient cytotoxic killing. Then, we explored the outcome of CD2 stimulation in CD8⁺ T cells by phosphoproteomics analysis and observed that it modulated about 19% of the CD8⁺ T cell phosphoproteome and diverse signaling pathways. Further analysis of CD2-regulated signaling revealed that the activity of the metabolic regulator adenosine monophosphate (AMP)–activated protein kinase (AMPK) on CTL lysosomes was essential for granule polarization toward the MTOC, defining a previously uncharacterized functional role for AMPK signaling in CD8⁺ T cells.

RESULTS

The abundance of cell surface CD58 determines the susceptibility of human B cells to lysis by freshly isolated CD8⁺ T cells

We previously identified several B cell types, including B cells from patients with chronic lymphocytic leukemia (CLL), that are resistant to lysis by freshly isolated CD8⁺ T cells in vitro (6). We hypothesized that such resistance was determined by an altered expression of sur-

face receptor(s) on resistant B cells. To test this hypothesis, we compared the cell surface proteomes of resistant and susceptible B cells through an aminooxy-biotinylation protocol coupled with streptavidin-based protein pulldown (10). Label-free liquid chromatography–coupled mass spectrometry (LC-MS/MS) analysis enabled us to quantify 1157 putative cell surface proteins, of which 257 were differentially expressed between resistant and susceptible B cells (Fig. 1A and table S1). Among these proteins, we identified five costimulatory receptors that were highly abundant on the surface of susceptible B cells [CD58, CD30, signaling lymphocytic activation molecule family member 1 (SLAMF-1), SLAMF-7, and intercellular adhesion molecule–1 (ICAM-1)] and one inhibitory receptor, the leukocyte-associated

immunoglobulin-like receptor 1 (LAIR1), that was present on the surface of resistant B cells (Fig. 1A). We knocked out the expression of these proteins in lysis-susceptible and lysis-resistant B cell lines, MEC-1 and BJAB cells, respectively, by CRISPR-Cas9-based technology (fig. S1A) to assess whether freshly isolated CD8⁺ T cells (containing >15% perforin-positive CTLs) could kill such targets loaded with bacterial superantigens (SAGs) (Fig. 1B). We found that only the absence of CD58 affected the killing process, because the lysis of CD58^{-/-} MEC cells was decreased by $70.1 \pm 7.4\%$ compared to that of CD58^{+/+} targets (Fig. 1C, left, and fig. S1, B and C). This effect was recapitulated by a CD58-blocking antibody (Fig. 1D). The rescue of CD58 expression on CD58^{-/-} MEC cells led to the restoration of killing by CD8⁺ T cells, which further showed a direct correlation between the cell surface abundance of CD58 and B cell susceptibility to cytotoxic killing (Fig. 1E and fig. S2A).

To better evaluate the mechanism underlying the resistance of CD58^{-/-} MEC cells to lysis, we also analyzed CD8⁺ T cell degranulation, which occurs when CTLs interact with antigen-loaded targets. Degranulation of CD8⁺ T cells can be detected by tracking the translocation of lysosome-associated membrane protein 1 (LAMP1) to the T cell surface during granule release (11). In contrast with the inhibition of cytotoxic killing, the genetic knockout or antibody-mediated blockade of CD58 on MEC cells led only to a moderate decrease ($20.8 \pm 19.6\%$) in CD8⁺ T cell degranulation (Fig. 1C, right, and fig. S2, B and C). This was paralleled by no significant effect on the efficiency of conjugation between CD8⁺ T cells and CD58^{-/-} MEC cells (fig. S2D). These observations suggest that although the extent of interaction between CD58^{-/-} targets and CTLs was largely preserved, such an interaction led to dysfunctional CD8⁺ T cell degranulation, resulting in poor cytotoxic killing.

To confirm our findings in a more physiologically relevant context of CTL–target cell interaction, we transfected freshly isolated CD8⁺ T cells to express a recombinant TCR (rTCR) specific for the Epstein-Barr virus (EBV)-derived peptide GLCTLVAML and assessed their cytotoxicity against peptide- or SAg-loaded CD58^{+/+} and CD58^{-/-} HLA-A2-overexpressing MEC cells (Fig. 1, F and G). In both settings, the lack of CD58 on the target cells led to the inhibition of CD8⁺ T cell–mediated killing, which was paralleled by dysfunctional degranulation (Fig. 1H). Together, these data suggest that the amount of CD58 on target B cells is a critical determinant for cytotoxic killing by freshly isolated CD8⁺ T cells.

CD2 costimulation promotes the polarization of lytic granules toward the MTOC at the immune synapse between perforin-positive CD8⁺ T cells and B cells

On the basis of our previous findings showing that defective killing of B cells is associated with the loss of granule recruitment to the CTL immune synapse (6), we hypothesized that the same mechanism might underlie the defect in the killing of CD58^{-/-} targets. Confocal imaging analysis revealed that, indeed, the absence of CD58 on target cells led to an impairment in lytic granule polarization to the MTOC docked at the immune synapse, whereas minimal effects were observed on the polarization of the MTOC itself (Fig. 2, A and B). This suggests that costimulation of CD8⁺ T cells by CD2 interacting with CD58 on B cells regulates granule convergence toward the MTOC. To test this hypothesis, we analyzed granule translocation in perforin-positive CD8⁺ T cells incubated with a combination of three anti-CD2 monoclonal antibodies that functionally mimic TCR engagement together with CD2-CD58-mediated costimulation (12–16) and ob-

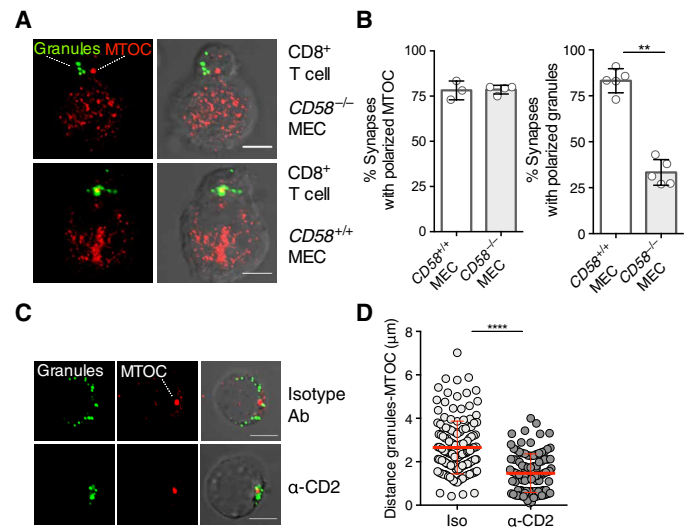


Fig. 2. CD2 engagement by CD58 is essential for the polarization of lytic granules to the immune synapse of freshly isolated CD8⁺ T cells. (A) Representative immunofluorescence images of CD8⁺ T cells incubated for 30 min with CD58^{+/+} and CD58^{-/-} MEC cells and stained with antibodies against perforin (to detect granules; green) and γ -tubulin (MTOC; red). Scale bars, 5 μ m. (B) Left: Quantitative analysis of MTOC polarization in perforin-positive CTLs incubated with the indicated MEC targets. Right: Quantitative analysis of lytic granule polarization in CD8⁺ T cells that exhibited MTOC docking at the immune synapse with MEC targets. Data are means $\pm$ SD of three to five experiments. Measurements were taken from 30 synapses for each condition. (C) Representative immunofluorescence images of CD8⁺ T cells incubated for 10 min with isotype control and CD2-specific (α -CD2) antibodies and stained with antibodies against perforin (granules) and γ -tubulin (MTOC). Scale bars, 5 μ m. (D) Quantitative analysis of lytic granule polarization in antibody-stimulated CD8⁺ T cells. Data are means $\pm$ SD of four experiments. Each dot represents individual granule. Data were analyzed by Mann-Whitney test. ** $P < 0.01$ and **** $P < 0.0001$.

served that CD8⁺ T cell stimulation through CD2 promoted granule polarization toward the MTOC (Fig. 2, C and D). Together, our data suggest that CD2 stimulation by either a physiological ligand or soluble stimuli promotes granule convergence toward the MTOC in freshly isolated CD8⁺ T cells, which, in the context of the immune synapse, results in complete polarization of T cells toward their targets.

Phosphoproteomics analysis of CD2 signaling reveals a functionally diversified signaling network of CD8⁺ T cell costimulation

To gain insights into the signaling pathways of the CD2 network that costimulate killing by CD8⁺ T cells and the formation of a lytic synapse and, in particular, those pathways implicated in the regulation of granule polarization, we performed quantitative, label-free proteomics analysis of cytoplasmic phosphorylation events in CD2-stimulated, enriched CD8⁺CD57⁺ T cells isolated from three individual donors. Analysis of merged samples labeled by tandem mass tags and phosphopeptide enrichment followed by LC-MS/MS revealed a total of 2920 phosphopeptides and 3470 phosphorylation events in 1373 proteins with a highly reproducible profile among donors (table S2 and fig. S3). We identified 549 phosphopeptides and 616 phosphorylation events in 373 proteins that were statistically significantly different between CD8⁺ T cells stimulated with anti-CD2 antibody and those incubated with isotype control antibody (Fig. 3 and table S3).

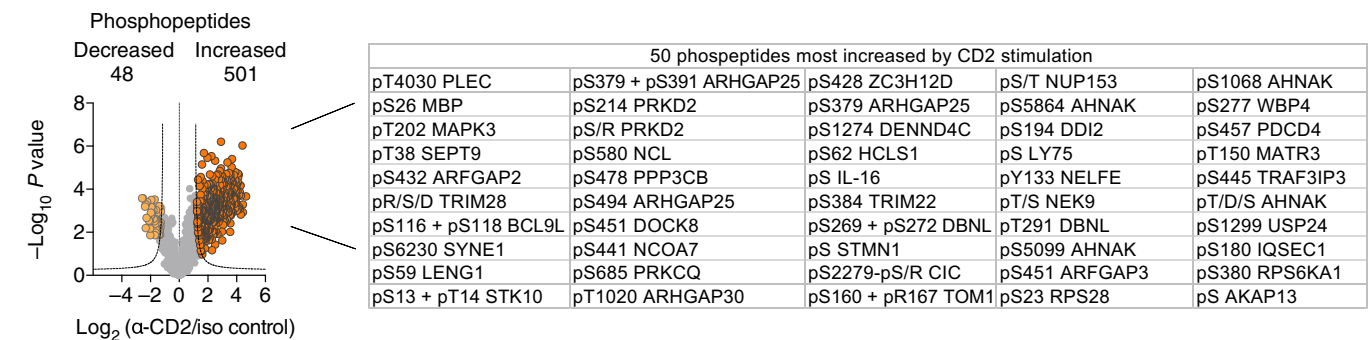


Fig. 3. CD2-regulated phosphoproteome in freshly isolated human CD8⁺ T cells. Left: Volcano plot shows the log₂ fold change in the abundance of phosphopeptides (CD2-stimulated/isotype control-treated) plotted against the $-\log_{10} P$ value highlighting phosphopeptides that statistically significantly decreased (light orange) or increased (dark orange). The FDR was 0.01. Data are from cells from three donors. Right: The 50 phosphopeptides that were most increased in abundance in response to CD2 stimulation, according to the fold change in CD2-stimulated versus isotype-stimulated CD8⁺ T cells.

The functional diversity of CD2-regulated phosphoproteins was assessed by analyzing the phosphoproteomics data with Cytoscape software, which enabled the generation of phosphoprotein networks based on the interaction data extracted from the STRING database (17). Our analysis revealed that CD2-regulated phosphoproteins formed a complex network of signaling events endowed with several features (fig. S4). First, CD2 regulated the phosphorylation status of a number of phosphoproteins that were predicted to be CD2 interactors (Fig. 4A), including several experimentally validated CD2 interactors, such as CD2AP (18), PTPRC/CD45 (19), and SH3KBP1/CIN85 (20, 21). Furthermore, cluster enrichment analysis of the detected phosphoproteins indicated that several functional groups of CD2-regulated phosphoproteins were statistically significantly enriched after CD2 stimulation (Fig. 4B and table S4). Among the enriched signaling pathways, four major pathways included TCR signaling ($n = 17$ proteins), endocytosis ($n = 15$), regulation of the actin cytoskeleton ($n = 14$), and mammalian target of rapamycin (mTOR) signaling ($n = 7$). Several relevant functional groups were also enriched, including phosphoproteins implicated in the regulation of cell activation and proliferation (cell cycle: $n = 51$; positive regulation of transcription: $n = 38$), vesicle-mediated transport group ($n = 54$), locomotion ($n = 45$), cytoskeletal organization ($n = 45$), positive regulation of guanosine triphosphatase (GTPase) activity ($n = 44$), and autophagy (positive regulation: $n = 10$; autophagy: $n = 9$). Note that we observed a partial overlap of the CD2 signaling network with the previously characterized phosphoproteomes of the TCR and CD28 (table S5) (22, 23). Together, the integrative network of the CD2-regulated phospho-events highlights the ability of CD2 to modulate the phosphorylation status of proteins implicated in not only many cellular processes, with particular emphasis on TCR signaling, cell activation, vesicular trafficking, and cytoskeletal reorganization but also metabolic and autophagic processes (Fig. 4C).

The CD2-regulated kinome reveals concomitant activation of the mitogen-activated protein kinase and mTOR pathways

To generate a comprehensive map of the CD2-regulated kinome, we merged our phosphoproteomics results with data from Western blotting analysis and in silico prediction of the upstream kinases shaping the CD2-regulated phosphoproteome. First, phosphoproteomics and Western blotting analyses revealed that CD2 modulated the phosphorylation of 40 kinases (table S4), including three kinases

belonging to the mTOR signaling pathway, namely, mTOR (Ser²⁴⁸), ribosomal protein S6 kinase β -1 (p70-S6K) (Thr³⁸⁹), and AMP-activated kinase (AMPK) (Thr¹⁷²) (Fig. 5A). Next, we assigned the upstream kinases for CD2-regulated phosphosites with the NetworKIN and NetPhorest packages (24). Of 392 phosphosites, 156 were assigned by NetworKIN/NetPhorest to 41 kinase groups and 82 individual kinases (table S6). Among those, most phosphorylation events were predicted to be mediated by the protein kinase C (PKC) group (13.9% of phosphosites), the mitogen-activated protein kinase (MAPK) group (11.7%), the Akt group (9.9%), and the JNK (c-Jun N-terminal kinase) group (8.8%). In support of this prediction, PKC δ (Ser³⁰⁴), PKC θ (Ser⁶⁸⁵), extracellular signal-regulated kinase 2 (ERK2) (MAPK1) (Thr¹⁸⁵), and ERK1 (MAPK3) (Thr²⁰²) were found to be phosphorylated on their activatory residues in our phosphoproteomics analysis (table S2). Western blotting analysis further revealed that the phosphorylation of Akt1 (Ser⁴⁷⁴) and JNK1 (Thr¹⁸³/Tyr¹⁸⁵) on their respective activatory residues was enhanced after CD2 stimulation (Fig. 5B). With this information in hand, we generated an integrative CD2-regulated kinome map, which was enriched for the kinases of the MAPK and mTOR signaling pathways (Fig. 5C).

CD2-mediated AMPK activation is essential for granule polarization toward the MTOC

We took advantage of the CD2 phosphoproteomics network to search for signaling events that were responsible for the polarization of lytic granules toward the MTOC. To this end, we concentrated on CD2-regulated lysosome-associated signaling molecules (table S7). Among these, we found PRKAA1 (protein kinase AMP-activated catalytic subunit α 1), the α 1 catalytic subunit of AMPK, which is selectively expressed in human T cells (25) and may have a role in lysosomal trafficking based on a study connecting the metabolic status of the cell with lysosomal positioning (26). We found that CD2 stimulation enhanced PRKAA1 phosphorylation on the activatory residue Thr¹⁷² (Fig. 5A) and on the residue Ser⁵²³ (Fig. 4C), which was among the top 25% most abundant phosphopeptides in CD2-stimulated CD8⁺ T cells (table S2). In addition, CD2 modulated the phosphorylation of validated AMPK-specific phosphosites (27) on MFF, GOLGA4, and PEA15 as detected by phosphoproteomics (Fig. 6A) and on RAPTOR1 and GSK3B as detected by Western blotting analysis (Fig. 6B), confirming that CD2 stimulation promoted AMPK activation. We also observed that, after receptor stimulation on CD8⁺ T cells, increased AMPK activation correlated with an enhanced

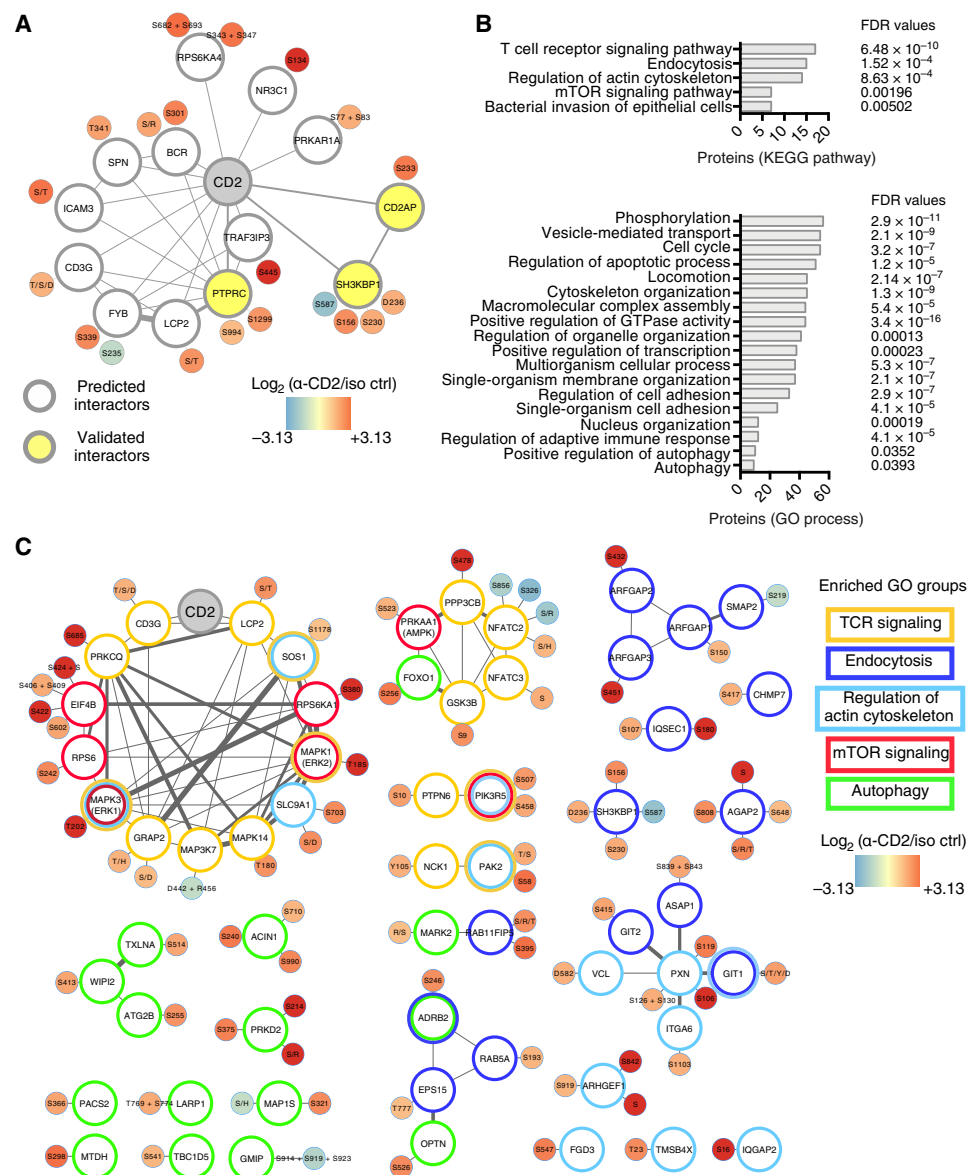


Fig. 4. Functional analysis of the CD2-regulated phosphoproteome in freshly isolated human CD8⁺ T cells. (A) A network of predicted and validated CD2-interacting proteins, the phosphorylation status of which was regulated by CD2 stimulation. The network was generated by the STRING application in Cytoscape. (B) Cluster enrichment analysis of CD2-regulated phosphoproteins (Cytoscape application; redundancy cutoff: 0.2). FDR *P* values are shown to the right. GO: Gene Ontology; KEGG, Kyoto Encyclopedia of Gene and Genomes. (C) An integrative network of the CD2-regulated phosphoproteins belonging to the indicated enriched functional groups was generated with the STRING application (confidence threshold, 0.4) and ClusterMaker application (confidence threshold, 0.641) in Cytoscape. For (A) and (C), phosphosite nodes are color-coded on the basis of the $\log_2$ fold change in phosphopeptide abundance (CD2-stimulated/isotype-stimulated). Edge width is proportional to the confidence of the interaction score based on the experimentally validated data from the STRING database.

ability of CD2 to promote granule polarization (fig. S5). Together, these findings motivated our decision to investigate the role of AMPK in granule trafficking.

To assess whether AMPK activity was essential for CD2-dependent granule polarization, we used CRISPR-Cas9 ribonucleoprotein (RNP) complex-mediated gene knockout to reduce AMPK abundance in freshly isolated CD8⁺ T cells ($52.9 \pm 15.2\%$ decrease; Fig. 6C and fig. S6).

This resulted in the impairment of CD2-stimulated granule mobilization toward the MTOC (Fig. 6D). These results were confirmed in experiments using a small interfering RNA (siRNA)-based approach (fig. S7), collectively showing that the ability of CD2 to induce optimal granule polarization was dependent on AMPK.

To assess whether AMPK activation was sufficient to induce granule polarization in the absence of CD2 stimulation, we performed a time course analysis of granule distribution in perforin-positive CD8⁺ T cells transfected with mRNA encoding a constitutively active AMPK variant (AMPK-CA) or a kinase-deficient variant (AMPK-KD). This analysis revealed that AMPK-CA, but not AMPK-KD, promoted granule clustering around the MTOC (Fig. 6E). Accordingly, T cell treatment with the AMPK-activating compounds A-769662 and metformin induced the polarization of lytic granules toward the MTOC (Fig. 6, F and G, and fig. S8). Together, these observations indicate that AMPK activation stimulates polarized granule recruitment to the MTOC in freshly isolated CD8⁺ T cells.

Lytic granule polarization is dependent on AMPK localized on lysosomes, but not on lytic granules themselves

Cytoplasmic AMPK accumulates at various subcellular compartments (28). We observed that in freshly isolated CD8⁺ T cells, $49.0 \pm 18.3\%$ of the intracellular AMPK pool that was detected by immunofluorescence colocalized with LAMP1-positive lysosomes (Fig. 7, A and B). To investigate whether the active AMPK that was responsible for granule movement was restricted to the lysosomal compartment, we transfected CD8⁺ T cells with constructs encoding the AMPK inhibitor peptide (AIP) to inhibit AMPK specifically at lysosomes (LAMP1-AIP-mCherry) and at mitochondria (Tom20-AIP-mCherry), the latter being a control (29). This analysis revealed that lysosome-specific, but not mitochondria-specific, inhibition of AMPK blocked CD2-induced

granule polarization in CD8⁺ T cells (Fig. 7, C and D), thus suggesting that lytic granule polarization is controlled by lysosome-restricted AMPK activity.

Furthermore, lysosome-associated AMPK did not colocalize with lytic granules themselves, although the two compartments were in close proximity to each other (Fig. 7, E and F). These data were confirmed by correlative light and electron microscopy (CLEM) analysis,

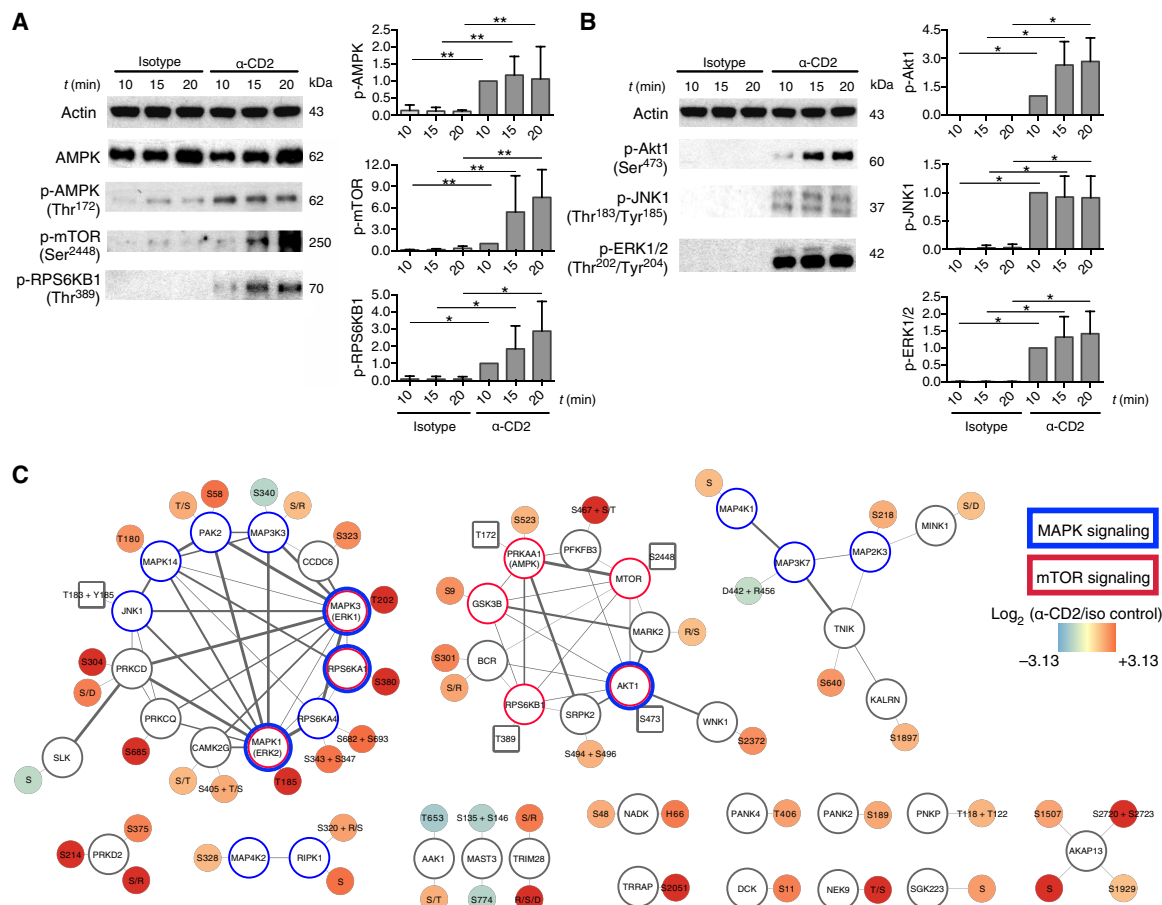


Fig. 5. CD2-regulated kinome in freshly isolated human CD8⁺ T cells. (A and B) Western blotting analysis of components of the mTOR signaling pathway (A) and the Akt/MAPK signaling pathway (B) in CD8⁺ T cells incubated with either isotype control antibody or CD2-specific antibody. Western blots are representative. Bar graphs show the normalized relative intensities of the indicated bands. Data are means $\pm$ SD of three or four experiments. Data were analyzed by Mann-Whitney test. * $P < 0.01$ and ** $P < 0.01$. (C) An integrative network of the CD2-regulated kinases was generated with the STRING application (confidence threshold, 0.4) and ClusterMaker (confidence threshold, 0.641) in Cytoscape. Round phosphosite nodes refer to the phosphoproteomics data and are color-coded on the basis of the log₂ fold change in phosphopeptide abundance (CD2-stimulated/isotype-stimulated). Square phosphosite nodes were retrieved from the Western blotting analysis. Edge width is proportional to the confidence of the interaction score based on the experimentally validated data from the STRING database.

which showed that AMPK-positive vesicles adjacent to lytic granules represented a separate vesicular entity (Fig. 7G). Together, these observations suggest a functional cross-talk between lytic granules and AMPK-positive lysosomes, whereby local AMPK activity is responsible for granule polarization.

DISCUSSION

In this study, in which we investigated CD2 signaling in human CD8⁺ T cells, we made the following four findings. First, we provided evidence for the essential role of CD2 in the formation of lytic immune synapses with B cells, through the costimulation necessary to promote efficient polarization of lytic granules and cytotoxic killing by freshly isolated CD8⁺ T cells. Second, we generated a map of the CD2 signaling network, which encompassed signaling pathways that regulate cell polarity, vesicular trafficking, cytoskeletal organization, immune responses, and metabolic processes. Third, we identified AMPK as a functionally critical node of the CD2 signaling network responsible for CD2-dependent granule polarization in perforin-

positive CD8⁺ T cells. Last, we showed that the AMPK pool that regulated this process was found on CTL lysosomes, and not on the granules themselves, thus illustrating previously uncharacterized functional cross-talk between distinct vesicular compartments in CD8⁺ T cells. A previous study described the interaction of lytic granules with the Rab27-positive late endosomal compartment, which is required for granule secretion (30).

CD2 has long been known as a potent costimulatory receptor for T cells (31), including CD8⁺ T cells (32, 33), and for another class of cytotoxic lymphocytes, natural killer cells (34, 35). The functional importance of CD2 costimulation applies to pathogen-specific T cell immunity (36) and autoimmunity (37, 38). High CD2 abundance selectively distinguishes memory T cells; hence, antibody-mediated ablation of CD2-high T cells has proven effective in counteracting autoreactive T cells in diabetes (39), psoriasis (40), and graft rejection in the clinic (41). In addition, CD2 costimulation appears to have an important role in antitumor immunity in B cell malignancies, because the loss of CD58 on tumor B cells is implicated in enhanced tumorigenesis in diffuse large B cell lymphoma, non-Hodgkin's B cell lymphoma,

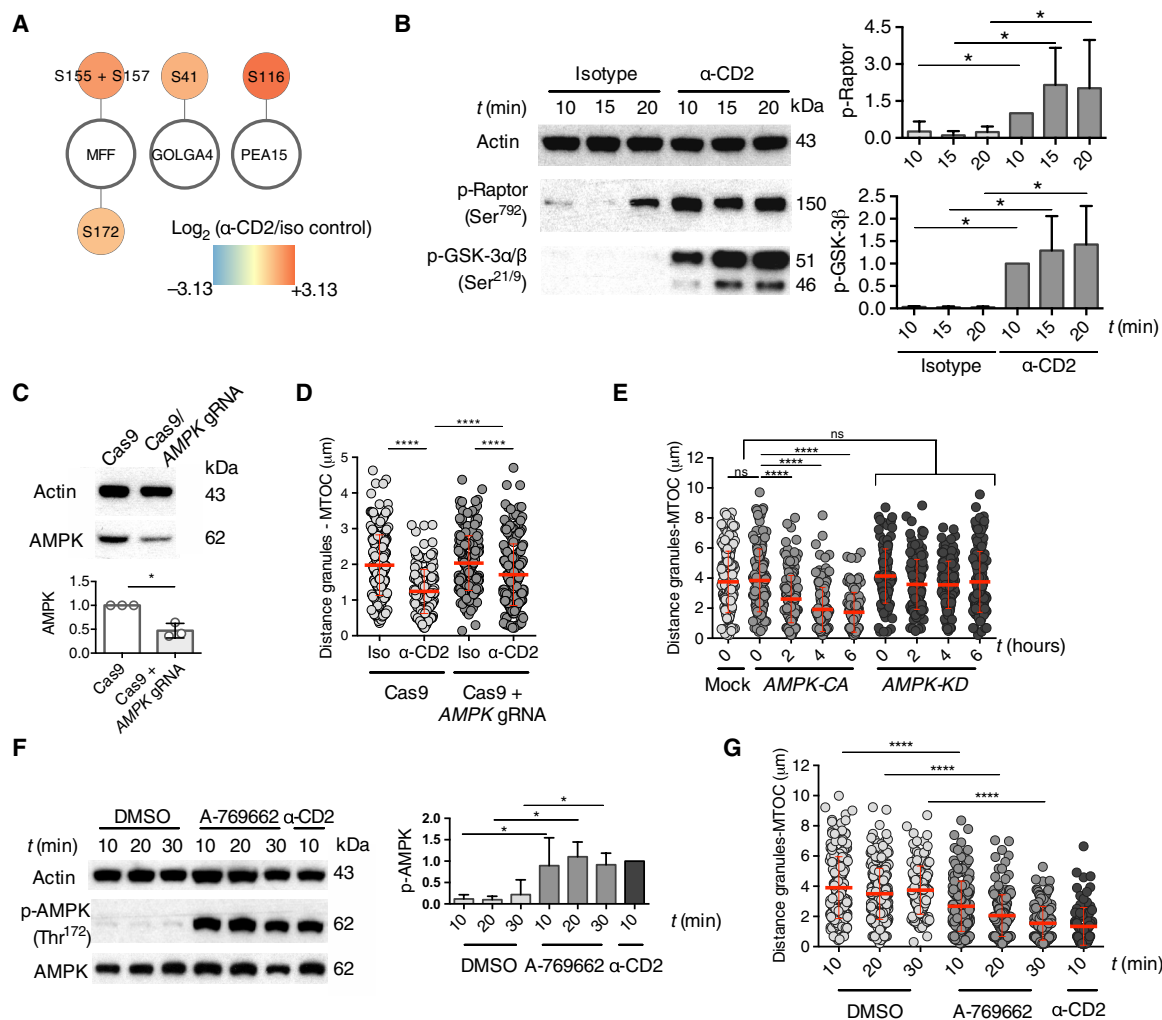


Fig. 6. CD2-mediated activation of AMPK is essential for granule polarization toward the MTOC. (A) CD2-regulated AMPK targets identified by phosphoproteomics. Phosphosite nodes are color-coded on the basis of the $\log_2$ fold change in phosphopeptide abundance (CD2-stimulated/isotype-stimulated). (B) CD2-regulated AMPK targets identified by Western blotting analysis of antibody-stimulated CD8⁺ T cells. Bar graphs show the normalized relative intensities of the indicated bands. Data are means $\pm$ SD of three or four experiments. (C) Western blotting analysis of CD8⁺ T cells electroporated with RNPs of Cas9 and AMPK-targeting gRNAs at 48 hours after electroporation. Bar graph shows the normalized relative intensities of the AMPK band. Data are means $\pm$ SD of three experiments. (D) Quantitative immunofluorescence analysis of granule polarization in the indicated Cas9-RNP-electroporated CD8⁺ T cells after incubation with isotype control or α CD2 antibody. Data are means $\pm$ SD of three experiments. (E) Quantitative immunofluorescence analysis of granule polarization in CD8⁺ T cells that were mock-electroporated or were electroporated with mRNA encoding the AMPK-CA or AMPK-KD constructs. Imaging analysis was performed at the indicated times after electroporation. Data are means $\pm$ SD of three experiments. (F) Left: Western blotting analysis of total and phosphorylated AMPK (a marker of activation) in CD8⁺ T cells treated with vehicle [dimethyl sulfoxide (DMSO)], 300 μ M A-769662, or CD2-specific antibody. Right: Bar graph shows the normalized relative intensities of the pAMPK-Thr¹⁷² band. Data are means $\pm$ SD of three experiments. (G) Quantitative immunofluorescence analysis of granule polarization in CD8⁺ T cells treated with vehicle (DMSO), 300 μ M A-769662, or CD2-specific antibody. Data are means $\pm$ SD of three experiments. Data were analyzed by Mann-Whitney test [for (B), (C), and (F)] or Kruskal-Wallis test [for (D), (E), and (G)]. * P < 0.05 and **** P < 0.0001; ns, not significant.

primary mediastinal large B cell lymphoma, and CLL (42–45). Our results suggest a molecular mechanism that may underlie these findings, because we showed that high CD58 amounts on B cells were required for efficient cytotoxic killing by human CD8⁺ T cells.

The CD2 receptor has been described as an adhesion and co-stimulatory receptor on T cells (33, 46, 47); however, signaling pathways explaining the potent effect of CD2 costimulation have been elucidated only partially. Known participants in the CD2 signaling network include the TCR CD3 ζ subunit (15, 48, 49), the kinases Lck (50) and Fyn (51), the transmembrane adaptor protein LAT (52, 53), and WASP (54). The CD2-dependent activation of Fyn has been

further linked to the activation of the PLC- γ 1/Vav1/PKC/Dok/FAK/Pyk2/JNK1 axis (51). Here, we characterized the CD2 signaling network by phosphoproteomics with the principal aim to elucidate signaling molecules aiding the TCR in the formation of functional immune synapses. First, our analysis revealed the existence of a broad CD2 signaling network in freshly isolated CD8⁺ T cells, which showed a substantial overlap with the published signaling phosphoprotein network controlled by the TCR (table S5) (23), in agreement with the data on TCR transactivation by CD2 previously reported by others (15, 48). Second, we observed that CD2 engagement on CD8⁺ T cells activated several signaling networks that are

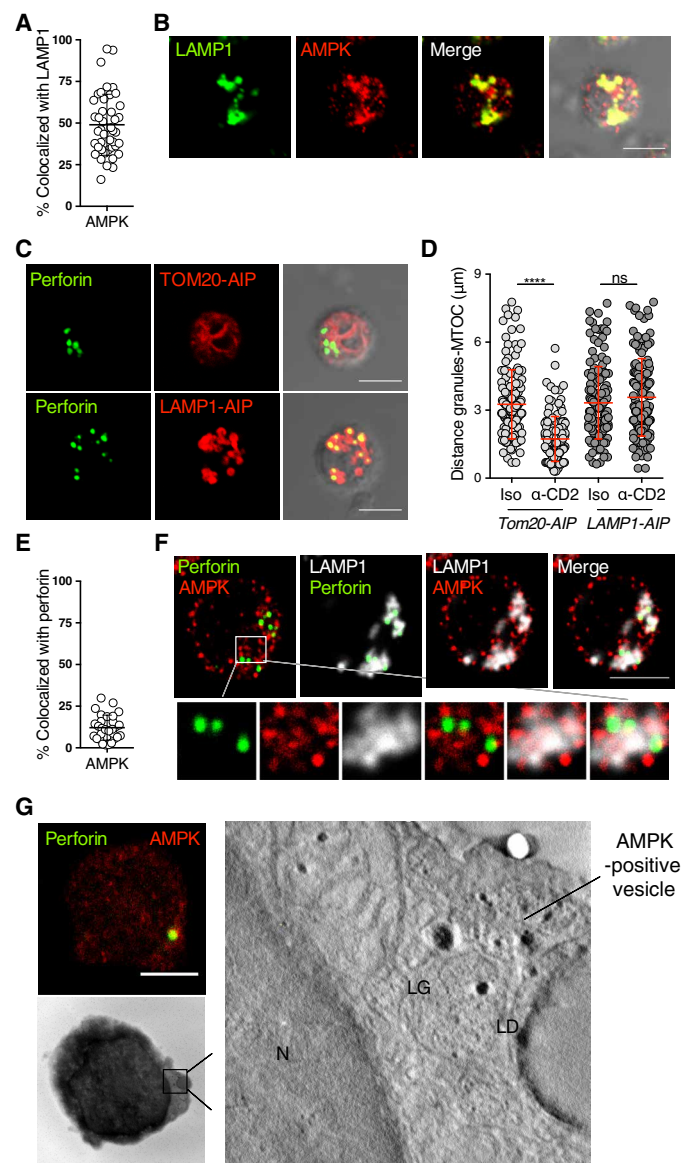


Fig. 7. Lytic granule polarization is driven by AMPK localized on CD8⁺ T cell lysosomes, but not on lytic granules themselves. (A and B) Colocalization analysis (A) and immunofluorescence imaging (B) of LAMP1 and AMPK signals in resting CD8⁺ T cells. Quantitative measurements were taken for 30 T cells from three donors. (C) Immunofluorescence images of resting perforin-positive CD8⁺ T cells electroporated with *LAMP1-AIP-mCherry* or *Tom20-AIP-mCherry* constructs and stained with antibodies against perforin (granules). (D) Quantitative analysis of CD2-induced lytic granule polarization in perforin-positive CD8⁺ T cells overexpressing either *LAMP1-AIP-mCherry* or *Tom20-AIP-mCherry* and treated with the indicated antibodies. Data are means $\pm$ SD of three experiments. (E) Colocalization analysis of immunofluorescence signals in resting CD8⁺ T cells stained with antibodies against AMPK and perforin (granules). Data are means $\pm$ SD of three experiments. (F) Immunofluorescence image of a representative resting CD8⁺ T cell stained with antibodies against AMPK, LAMP1, and perforin (granules). Images are representative of three experiments. (G) CLEM analysis of lytic granules and the AMPK-positive compartment in resting CD8⁺ T cells. N, nucleus; LD, lipid droplet; AMPK, AMPK-positive vesicle; LG, lytic granule. Scale bar, 5 μ m. Images are from a single representative experiment. Data in (D) were analyzed by Kruskal-Wallis test. **** P < 0.0001.

relevant for immune synapse assembly, namely, those regulating vesicular trafficking and cytoskeletal organization. Note that these particular functional groups of proteins were among the phosphoproteins that distinguished the CD2 phosphoprotein network from the previously characterized signaling phosphoproteomes of the TCR and CD28 (table S5) (22, 23), thus suggesting that CD2 signaling may play a specific role in organizing these processes in T cells. In addition, we found that CD2 regulated the phosphorylation of several proteins previously implicated in the polarization events occurring at the immune synapse of CTLs and natural killer cells, including paxillin (Ser¹⁰⁶, Ser¹¹⁹, Ser¹²⁶, and Ser¹³⁰) (55), Dock8 (Ser⁴⁵¹) (56), and PKC δ (Ser³⁰⁴) (table S3) (57).

We were particularly intrigued by our finding that AMPK represents a principal signaling node of the CD2 network responsible for the regulation of granule convergence toward the MTOC in freshly isolated CD8⁺ T cells. Previously, studies performed on CTLs expanded *in vitro* showed that granule polarization at the immune synapse requires high-affinity TCR-peptide-major histocompatibility complex interactions and rapid kinetics of intracellular Ca²⁺ flux (2, 4), without an apparent need for CD2 costimulation. Our work indicates instead that freshly isolated CD8⁺ T cells rely on CD2 costimulation for efficient granule polarization at the immune synapse with B cells, even in the presence of high-affinity TCR engagement by SAg. Because the absence of CD58-mediated CD2 signaling did not affect the ability of CD8⁺ T cells to form immune synapses and polarize the MTOC, we hence propose that in freshly isolated CD8⁺ T cells, TCR engagement is essential to promote these processes, whereas CD2-mediated costimulation, through its enhancement of AMPK activation, serves to ensure complete polarization of lytic granules to the MTOC.

At the molecular level, the involvement of AMPK in lytic granule trafficking in CD8⁺ T cells raises two questions related to the regulation of this process. First, our findings illustrate an example of a direct cross-talk between metabolic signaling and the regulation of immune synapse formation. AMPK is the metabolic regulator that, in CD8⁺ T cells, controls the development of immune responses to infections and the formation of immunological memory (58, 59). In the context of antitumor immunity, the ability of AMPK to promote granule convergence to the MTOC makes it plausible to propose that AMPK-activating drugs might facilitate the formation of functional lytic immune synapses whenever granule trafficking is affected by tumor immune evasion. In support of this, we observed that treatment with the AMPK-activating antidiabetic compound metformin, which is known for its antitumor activity (60), promoted granule polarization in CD8⁺ T cells (fig. S8). The antitumor action of metformin was previously shown to be dependent on intratumoral CTLs, in which metformin counteracts the development of an “exhausted” immunophenotype (61, 62). Our results reinforce these findings by suggesting that this mechanism of metformin action may be aided by the metformin-mediated enhancement of T cell polarization at the immune synapse.

Second, an important question is the characterization of the molecular events that enable AMPK activation to result in the polarized movement of granules toward the MTOC. Changes in the metabolic status of the cell have been implicated in the movement of “conventional” lysosomes, because cell starvation causes lysosome polarization toward the MTOC, which is toward the minus ends of microtubules (26). Driven by the hypothesis that lysosomal positioning could be controlled by the starvation-induced activation of AMPK, our data

suggest that the activation of AMPK may cause the minus-end translocation of lytic granules in CTLs. The underlying molecular mechanism is as yet elusive. We previously reported that granule positioning at the periphery of CTLs is promoted by the Arf-like GTPase Arl8, which is the known mediator of the plus-end-directed transport of lysosomes (6). When Arl8 expression is reduced, lytic granules collapse at the MTOC (6), which suggests that the constitutive activity of Arl8 opposes the constant minus-end-directed pulling force and thus keeps lytic granules dispersed. In this context, we hypothesize that granule polarization may be induced by the CD2-dependent inhibition of Arl8 activity. Activated AMPK was shown to associate with the LAMTOR/Ragulator complex, which acts as a negative regulator of Arl8 (63, 64). We hence suggest that the involvement of CD2-dependent AMPK activation in Ragulator-dependent inhibition of Arl8 activity should be investigated in future studies.

Here, we illustrated an essential role for CD2 and AMPK signaling in the orchestration of lytic immune synapse assembly with B cell targets, implicating this axis in immune evasion in B cell malignancies. Our data further suggest that CD2 signaling may have a broader function in CD8⁺ T cells, because it modulates signaling events participating in the regulation of cellular metabolism and autophagy, which have emerged as critical regulators of CD8⁺ T cell differentiation and memory formation (65–67). Note that CD2 has the ability to activate two antagonistic, metabolism-linked signaling pathways. On the one hand, CD2 promotes mTOR/p70-S6K/Akt1 signaling, which activates anabolic processes. On the other hand, CD2 activates AMPK, which is a key regulator of catabolic processes, including autophagy. It was suggested that the concurrent activation of mTOR and AMPK signaling serves to sustain the rapid induction of biosynthetic processes through autophagy-dependent clearance of misfolded proteins (68) and to achieve high rates of protein secretion (69). Our data thus suggest that CD2 signaling may be directly implicated in the shaping of metabolic, autophagic, and secretory profiles in CTLs. A better understanding of this regulation should constitute a basis for the rational manipulation of CD2 costimulation in human pathologies, including hematological disorders.

MATERIALS AND METHODS

Lymphocyte isolation and culture

The collection of peripheral blood samples from healthy donors was approved by the review board and performed after receiving signed informed consent according to institutional guidelines. CD8⁺ T cells were purified from peripheral blood (with at least 15% of perforin-positive cells among the total CD8⁺ cells; average, $23.0 \pm 12.3\%$) by negative selection with RosetteSep cocktails (STEMCELL Technologies) to >90% purity. The cells were then cultured in complete RPMI 1640-Hepes [7.5% iron-enriched HyClone fetal calf serum (FCS), 2 mM L-glutamine and penicillin (50 IU/ml)] and were either used immediately for assays or frozen upon isolation and then used after 40 to 48 hours of recovery in complete RPMI 1640-Hepes containing interleukin-15 (IL-15) (0.01 ng/ml; Miltenyi).

B cell cultures

The CLL-derived B cell line MEC-1 (MEC) (70) and EBV-transformed B cells (EBV-B) were cultured in complete RPMI 1640 with 7.5% iron-enriched HyClone FCS (Thermo Fisher Scientific), 2 mM L-glutamine, and penicillin (50 IU/ml). The Burkitt lymphoma-

derived B cell line BJAB was cultured in complete RPMI 1640 medium supplemented with 1 mM pyruvate.

Reagents

Reagents and antibodies used for Western blotting, flow cytometry, and immunofluorescence are listed in tables S8 to S10. Staphylococcal SAgS (staphylococcal enterotoxins SEA AT101, SEB BT202, and SEE ET404) were purchased from Toxin Technology. Bovine serum albumin (BSA), poly-L-lysine, propidium iodide, and saponin were obtained from Sigma-Aldrich. Monensin was from BioLegend. A-769662 and metformin hydrochloride were from Cayman Chemical. Proteomics reagents (periodate, aminoxy-biotin, aniline, lauryl maltoside, and iodoacetamide) were from Pierce (Thermo Fisher Scientific).

CRISPR-Cas9 mutagenesis of B cells and CD8⁺ T cells

Guide RNAs (gRNAs) were designed with the CRISPOR webtool (71) for *CD58* (GAGCATTACAACAGCCATCG), *SLAMF-1* (CGATCTCCTAGATAACGTGG), *SLAMF-7* (GAGCTGGTTCG-GTTCCGTTGG), *CD30* (CGGGTCGACATTTCGACAGACA), *ICAM1* (dual-combination TTAAGTGCACACGTCAGCCGC and CGTGATTCTGACGAAGCCAG), *LAIR-1* (TACTGAGTCAATG-CGGAATC), and *AMPK* (dual-combination AAGATCGGCCAC-TACATTCT and ATTCGGAGCCTTGATGTGGT). gRNA sequences for AMPK are shown in fig. S6. gRNAs for B cell mutagenesis were cloned into pSpCas9(BB)-2A-GFP (PX458) plasmid (a gift from F. Zhang; Addgene, 48138) as previously described (72). B cells were nucleofected with gRNA-encoding vectors or control empty vector using the electroporation buffer V [90 mM Na₂HPO₄, 90 mM NaH₂PO₄, 5 mM KCl, 10 mM MgCl₂, and 10 mM sodium succinate (pH 7.2)] and program W-003 with an Amaxa Nucleofector II system (Lonza) (73). Green fluorescent protein (GFP)-expressing cells were sorted, subcloned, and screened for gene knockdown by flow cytometry staining. For T cell treatment, 5×10^6 CD8⁺ T cells were resuspended in 100 μ l of Amaxa T cell nucleofection buffer (Lonza) and nucleofected by V-024 pulse with Cas9 RNPs prepared by incubating 5 μ g of Cas9 [Integrated DNA Technologies (IDT)] with 3 μ g of in vitro-transcribed gRNA prepared as described previously (74). Briefly, DNA sequences corresponding to gRNAs were amplified using the PX458 plasmid as a template, the universal reverse primer AGCACCGACTCGGTGCCACT, and the following forward primers: (i) TTAATACGACTCACTATAGGAAGATCGGCCACTACATTCT-gtttagagctagaatagc and (ii) TTAATACGACTCACTATAGGAT-TCGGAGCCTTGATGTGGTgtttagagctagaatagc. After electroporation, T cells were allowed to recover in RPMI 1640-Hepes containing 20% FCS for 48 hours and then were used for functional assays.

Flow cytometry assays of cytotoxicity and degranulation

These assays were performed as described previously (6). Briefly, B cells were loaded with a mixture of SEA (1 μ g/ml), SEB, and SEE; EBV BMFL1_(280–288) peptide (20 μ g/ml; IBA GmbH), or 1% BSA for controls. For the cytotoxicity assay, B cells (25×10^3) were incubated with carboxyfluorescein diacetate succinimidyl ester (CFSE)-loaded CD8⁺ T cells at the ratio indicated in the figure legend in 50 μ l of complete RPMI 1640-Hepes for 4 hours at 37°C. Cytotoxicity was calculated as follows: (% CFSE⁺ propidium⁺ dead cells – % CFSE⁺ propidium⁺ dead cells in control sample) $\times$ 100/(100 – % CFSE⁺ propidium⁺ dead cells in control sample). Mean background levels of cytotoxic killing were $3.2 \pm 0.6\%$. For degranulation assays, APCs

were incubated with CFSE-loaded CD8⁺ T cells in 50 μ l of complete RPMI 1640-Hepes with LAMP1-specific antibody (1.2 μ g/ml; H4A3, BioLegend) for 1 hour at 37°C before 25 μ l of monensin (1:333) was added for another 3 hours. Fixed cells were permeabilized and stained with Alexa Fluor 647–conjugated anti-mouse immunoglobulin G (H+L) antibody (Invitrogen) diluted in phosphate-buffered saline (PBS), 0.2% saponin, and 1% FCS. The extent of T cell degranulation was calculated as follows: (% CFSE⁺LAMP1⁺ cells – % CFSE⁺LAMP1⁺ cells in control sample) $\times$ 100/(100 – % CFSE⁺LAMP1⁺ cells in control sample). Mean background levels of degranulation were $1.9 \pm 1.1\%$.

Immunofluorescence microscopy

For immune synapse analysis, 50×10^3 APCs were loaded with SAG mixture and then washed in RPMI 1640-Hepes and incubated with CD8⁺ T cells (150×10^3) in 20 μ l of RPMI 1640-Hepes in a 37°C bath for the times indicated in the figure legend. For the past 15 min of incubation, the cells were allowed to adhere to the imaging slides and then were fixed (75) and stained as indicated. For granule polarization analysis, CD8⁺ T cells (150×10^3) were left to rest in complete RPMI 1640-Hepes for 10 min and then were incubated with anti-CD2 antibodies, 300 μ M A-769662, or 3 mM metformin in a final volume of 40 μ l. At the end of the stimulation, the cells were washed with ice-cold RPMI 1640-Hepes, fixed onto slides, and stained as indicated. Confocal microscopy on 0.9- μ m-thick sections was performed on an LSM 700 (Carl Zeiss) or Leica TCS SP5 with a 63 $\times$ objective lens, and captured images were processed with ImageJ software. The distance between lytic granules and MTOC was calculated in at least 30 individual T cells per experiment and condition with ImageJ software. Colocalization analysis was performed by calculating the Manders' coefficient with the JACoP plugin in ImageJ.

Western blotting

CD8⁺ T cells (1.5×10^6) were stimulated or treated with the compounds indicated for immunofluorescence analysis and then lysed in 40 μ l of lysis buffer [20 mM Tris (pH 8.0), 165 mM NaCl, 5 mM EDTA, 1% NP-40, supplemented with protease inhibitors (Calbiochem), 1 mM Na₃VO₄, and 25 mM NaF]. Equal amounts of postnuclear cell lysates (10 to 20 μ g) were analyzed by Western blotting with the appropriate primary antibodies, and band intensities on scanned images were quantified with ImageJ software.

Validation of AMPK-specific gRNAs with in vitro–expanded CD8⁺ T cells

CTL blasts were generated from freshly isolated CD8⁺ T cells by incubation with CD3/CD28 Dynabeads (Thermo Fisher Scientific). CTLs were cultured in complete RPMI 1640 supplemented with IL-2 (10 ng/ml; Miltenyi), glutamine, penicillin/streptomycin, and 10% FCS. Forty-eight hours after the beginning of expansion, CTLs were depleted of beads, washed once with PBS, and resuspended in electroporation buffer 1M [5 mM KCl, 15 mM MgCl₂, 120 mM Na₂HPO₄/NaH₂PO₄ (pH 7.2), and 50 mM Mannitol; 100 μ l for 2×10^6 CTLs] (76). In parallel, Cas9 RNP complexes were prepared by incubating 5 μ g of Cas9 (IDT, 1 μ l) with 3 μ g of each gRNA. After 20 min of incubation at room temperature, RNPs were mixed and used to nucleofect CTL blasts by applying pulse V-024 with an Amaxa Nucleofector II (Lonza). CTLs were allowed to recover in complete RPMI 1640 with 20% FCS and without antibiotics overnight and then were cultured as described earlier. AMPK knockout was assessed by Western blotting analysis 72 hours after nucleofection.

CD8⁺ T cell nucleofection with siRNA

Freshly isolated CD8⁺ T cells (5 to 7×10^6) were nucleofected with 100 ng of AMPK-specific siRNA pool (EHU074041, Sigma-Aldrich) or GFP-specific siRNA control (EHUEGFP) in 100 μ l of Amaxa T cell nucleofection buffer (Lonza) and left to recover in RPMI 1640-Hepes, 10% FCS for 40 hours before the extent of protein knockdown was assessed.

mRNA production and nucleofection of B cells and CD8⁺ T cells

The complementary DNA (cDNA) encoding full-length CD58 was amplified using MEC cDNA as a template (forward primer: CTAGCTAGCACCATTGGTTGCTGGGAGC; reverse primer: CCGCTCGAGTCAATTGGAGTTGGTTCTGTC) and subcloned into pcDNA3.1(+). The cDNA encoding full-length HLA-A2 (human leukocyte antigen A*02) was synthesized on the basis of the sequence provided by J. Riley (University of Pennsylvania) and subcloned into pcDNA3.1(+) between the Eco RV and Bam HI sites. EBV-reactive rTCRs were isolated from a T cell line raised by repetitive stimulation with the HLA-A2–restricted, EBV-derived peptide BMLF1280-288 GLCTLVAML. Using high-precision 96-well sorting on an Aria Fusion cell sorter (BD Biosciences), viable CD3⁺CD8⁺ EBV-reactive T cells were placed at one cell per well into 4.45 μ l of lysis buffer. Buffer preparation, polymerase chain reaction amplification, and sequencing of TCR α and β chains were performed as described previously (77). Multiple recurrent TCR pairs were selected for cloning and functional testing. TCR α and β chain V(D)J sequences that were codon-optimized for human expression were synthesized individually (Eurofins) and introduced using a seamless cloning approach into pcDNA3.1-TRAC and pcDNA3.1-TRBC vectors containing the murine α and β constant regions with an additional disulfide bond (78, 79). BMLF1 reactivity of rTCR was confirmed by exposing rTCR-transduced T cells to targets pulsed with BMLF1280-288 or an irrelevant HLA-A2–restricted peptide. The cDNA encoding AMPK-CA was amplified using pCIP-AMPKa1_WT as a template (a gift from R. Shaw; Addgene, plasmid no. 79010) (80) with the forward primer CCGGCTAGCACCattggc-gacagccga and the reverse primer CCGCTCGAGTTAGTAAAGACAGCTGAGAACTTC and subcloned into pcDNA3.1(+). The cDNA encoding AMPK-KD was amplified using pCIP-AMPKa1_KD as a template (Addgene, plasmid no. 79011) (80) with the forward primer CCGGCTAGCACCattggc-gacagccga and the reverse primer CCGCTCGAGTTATTGTGCAAGAATTTTAATTAGA and subcloned into pcDNA3.1(+). For overexpression studies, mRNA was prepared in vitro with 1 μ g of Xba I–linearized pcDNA3.1-based construct using HiScribe T7 ARCA mRNA Kit with tailing (New England Biolabs). MEC cells were nucleofected with 5 μ g of CD58- or HLA-A2–encoding mRNA using buffer V and program W-003. CD8⁺ T cells were nucleofected with the rTCR α and β pair (5 μ g of each mRNA) or with 5 μ g of AMPK-CA/KD mRNA using homemade buffer 1M and program V-024 with an Amaxa Nucleofector II (Lonza). After nucleofection, cells were maintained in complete RPMI 1640 (with 10% FCS for T cells) without antibiotics. Overexpression of recombinant proteins was verified by flow cytometry analysis with CD58-, HLA-A2-, and anti-mouse TCR β chain-specific antibodies (to detect rTCR).

CD8⁺ T cell nucleofection with LAMP1-AIP/Tom20-AIP plasmids

For overexpression of LAMP1-AIP and Tom20-AIP, CD8⁺ T cells were nucleofected with 2 μ g of purified LAMP1-mChF-AIP– and

Tom20-mChF-AIP-expressing plasmids (a gift from T. Inoue; Addgene, plasmid nos. 61524 and 61512).

Proteomics

Enrichment of cell surface proteins from B cells was performed as described previously (10). Briefly, biological duplicates of 120 to 160×10^6 B cells (MEC, EBV-B, BJAB, and B cells from patients with CLL) were washed twice in ice-cold PBS and incubated on rocking platform in 3 to 4 ml of freshly prepared biotinylation/oxidation solution (1 mM periodate, 100 μ M aminooxy-biotin, and 10 mM aniline) for 30 min at 4°C. The cell suspension was then quenched with 1% glycerol. Cell pellets were washed in PBS, 5% FCS, and in PBS containing CaCl_2 and MgCl_2 and then were lysed in 3 to 4 ml of 0.5% lauryl maltoside lysis buffer [150 mM NaCl and 10 mM tris-HCl (pH 7.6)]. Lysates were supplemented with 5 mM iodoacetamide with protease inhibitors (Roche) and left shaking for 30 min at 4°C. Subsequently, lysates were centrifuged once at 2800g for 5 min and twice at 16,000g for 10 min to eliminate nuclei. Equal amounts of protein per sample were left shaking with high-affinity streptavidin agarose resin [Pierce (Thermo Fisher Scientific)] at 4°C for 2 hours. Streptavidin agarose was pelleted at 1000g for 1 min and washed using Snap Cap columns [Pierce (Thermo Fisher Scientific)] 20 times with lysis buffer and then 20 times with PBS/SDS buffer (0.5% SDS). Agarose pellets were reduced in PBS/SDS with 100 mM dithiothreitol (DTT) for 20 min at room temperature in the dark, washed three times with 6 M urea and 100 mM tris-HCl (pH 8.5), alkylated in 20 mM iodoacetamide for 20 min in the dark, and washed 17 times with UC buffer and 3 times with 50 mM ammonium bicarbonate (pH 8.0). Agarose beads were digested by incubation with trypsin (Promega) overnight. Peptides were desalted on reversed-phase cartridges (SOLA, Thermo Fisher Scientific) according to the manufacturer's instructions.

CD8⁺ T cell stimulation, tandem mass tag (TMT) labeling, hydrophilic interaction liquid chromatography (HILIC) fractionation, and immobilized metal affinity chromatography (IMAC) phosphopeptide enrichment

Freshly isolated human effector CD8⁺ T cells (15×10^6) from three donors (67.2, 50.5, and 45.7% effector CD57⁺CD8⁺ T cells, respectively) were stimulated for 10 min at 37°C in complete RPMI 1640-Hepes medium with isotype (10 μ g/ml) or a mixture of CD2-specific antibodies (T11.1, T11.2, and HIK27 clones, all from Sanquin) each at 5 μ g/ml. Cells were pelleted and lysed in 250 μ l of 1% NP-40 lysis buffer [150 mM NaCl, 5 mM EDTA, and 20 mM tris-HCl (pH 8.0)]. Protein (220 μ g) from postnuclear lysates was reduced in 5 mM DTT, alkylated in 20 mM iodoacetamide, and precipitated by methanol-chloroform extraction. Protein pellets were resuspended in 100 mM triethylammonium bicarbonate and digested with 4.4 μ g of trypsin (Promega) overnight at 37°C. Desalted peptides (SOLA cartridges, Thermo Fisher Scientific) were labeled with six channels of a TMT10plex (Thermo Fisher Scientific) according to the manufacturer's instructions, except that 220 μ g of total protein/0.8 mg of TMT reagent was used. Excess TMT was quenched with 40 mM tris-HCl. The labeled peptides were pooled, desalted on Sep-Pak Plus cartridges (Waters), and dried down in a vacuum centrifuge. Phosphopeptides were enriched by HILIC/IMAC as described previously (81). Briefly, peptides were prefractionated on a HILIC column (Amide 80, 4.6 mm by 250 mm, Tosoh Bioscience LLC). Thirty fractions were collected and concatenated into 10 pools. Fractions were each incubated with 30 μ l of a 50% slurry of PHOS-Select Iron affinity gel (Sigma-Aldrich) for 30 min

at room temperature. Samples were washed twice with 0.5 ml of 250 mM acetic acid/30% ACN and then with double-distilled water before being eluted with 100 μ l of 400 mM ammonium hydroxide.

Liquid chromatography–coupled mass spectrometry

Peptides from the surface proteome and phosphoproteomics analyses were separated on an EASY-Spray column ES803 and analyzed on a Dionex UltiMate 3000/Orbitrap Fusion Lumos platform (all from Thermo Fisher Scientific) (82). For TMT-labeled peptides, we used the MultiNotch MS3 method (83) to generate spectra for the identification and quantitation of phosphopeptides with parameters as deposited in the data acquisition method [PDX PRIDE; (84)].

Surface proteome data processing

We used label-free quantitation as available in Progenesis QI (Waters) for the differential quantitation of surface-enriched proteins. Raw data were imported into Progenesis QI using default parameters. Tandem MS (MS/MS) data were searched with Mascot (Matrix Science, v. 2.5) against the UPR human database (retrieved 7 September 2016). Mass tolerances were set to 10 parts per million (ppm) for precursor and 0.5 Da for fragment masses. Carbamidomethylation (Cys) was set as fixed, and deamidation (Asn and Gln) and oxidation (Met) were set as variable modifications. Peptide-level false discovery rate (FDR) was adjusted to 1%, and peptides with a score of <20 were discarded. Quantitative data were median-centered in Progenesis QI based on identified peptides with the thresholds described earlier and extracted for further data processing in Perseus.

Phosphoproteomics data processing

LC-MS/MS data were analyzed with Proteome Discoverer version 2.2. Proteins were identified with Sequest HT against the UPR human database (retrieved 31 January 2018). Mass tolerances were set to 10 ppm for precursor and 0.5 Da for fragment masses. TMT10plex (N-terminal, K) and phosphorylation (STY) were set as dynamic modifications, whereas alkylation (C) was set as a static modification.

Analysis of proteomics and phosphoproteomics data

Perseus software was used to perform statistical analysis of proteomics data to identify proteins and phosphopeptides that were present at statistically significantly different amounts in the compared samples ($\log_2$ fold change, ≥ 1.0). Statistical analysis was performed by Student's *t* test on $\log_2$ -transformed values using FDR curvatures indicated in the supplementary table legends. Before that, peptide lists were filtered to leave only the lines where at least one experimental group had three valid values to select for events with a highly reproducible profile among donors. The median of each column was subtracted from the $\log_2$ -transformed values, and the missing values were imputed from the normal distribution. The proteins identified as being regulated by CD2 stimulation were subjected to functional analysis with Cytoscape software, which was used to generate protein interaction networks using the STRING application (confidence threshold set at 0.4) and the ClusterMaker application (MCL cluster; granularity: 2.7, STRING score–based edge cutoff: 0.641). Cytoscape was also used to annotate the identified phosphoproteins with Gene Ontology and Kyoto Encyclopedia of Genes and Genomes pathway protein classification and to perform cluster enrichment analysis (redundancy cutoff set to 0.2). NetworKIN/NetPhorest webtools were used to map identified phosphorylation sites with kinase motifs. For compatibility with NetworKIN, the numbering of phosphorylated residues was

manually corrected to correspond to the first protein isoform because Proteome Discoverer reports phosphosite positions for all potential isoforms of the relevant protein. Therefore, in the figures and in tables S3 and S5, the numbering for the first protein isoform is shown.

Correlative light and electron microscopy

CD8⁺ T cells (3×10^6 cells) were fixed onto poly-L-lysine-coated 35-mm imaging dishes (MatTek), fixed in 4% paraformaldehyde and 0.05% glutaraldehyde for 10 min at room temperature, quenched with PBS and 0.2 M glycine, and stored in PBS. Fixed cells were permeabilized with 0.1% saponin and 0.1% BSA, blocked with 5% BSA, and stained with AMPK-specific and perforin-specific antibodies (see table S9 for concentrations). Cells of interest were then imaged with a Zeiss LSM 700 confocal microscope together with the image of the MatTek grid that was imaged in phase contrast. The cells were then processed for electron microscopy analysis. Briefly, the cells were stained with osmium tetroxide (1%) and potassium ferrocyanide (1.5%) to label the membranes, which was followed by staining with uranium acetate (0.5%). The cells were then dehydrated in a graded series of alcohol and embedded in Epon 812. The marking in the grid was used to locate the cell of interest, which was then sectioned and imaged in an FEI Tecnai 12 electron microscope.

Statistical analysis

Calculation of mean and SD values and statistical analysis (Mann-Whitney test; Kruskal-Wallis test with post hoc Dunn's multiple comparisons test) were performed with GraphPad Prism software.

SUPPLEMENTARY MATERIALS

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Fig. S1. Flow cytometric analysis of surface receptors on B cells.

Fig. S2. Influence of CD58 abundance on target cells on CD8⁺ T cell killing and degranulation.

Fig. S3. Pearson's correlation analysis of the fold change in CD2-regulated phosphopeptide abundances among the three donors.

Fig. S4. CD2-regulated phosphoproteome.

Fig. S5. Comparison of the protein phosphorylation events induced by CD2 and TCR stimulation.

Fig. S6. Validation of AMPK-targeting gRNAs in CTL blasts.

Fig. S7. AMPK knockdown impairs CD2-dependent granule polarization.

Fig. S8. Metformin induces the polarization of granules to the MTOC.

Table S1. B cell surface proteome analysis.

Table S2. Phosphoproteome of unstimulated and CD2-stimulated human CD8⁺ T cells.

Table S3. CD2-regulated phosphoproteome in human CD8⁺ T cells.

Table S4. Enriched functional protein groups in the CD2-regulated phosphoproteome.

Table S5. Comparison of the CD2-regulated and published TCR- and CD28-regulated phosphoproteomes.

Table S6. NetworkIN/NetPhorest-predicted kinases of the CD2-regulated phosphosites.

Table S7. CD2-regulated lysosome-associated proteins in human CD8⁺ T cells.

Table S8. Antibodies and reagents used for flow cytometry and cell stimulation.

Table S9. Antibodies and reagents used for immunofluorescence microscopy.

Table S10. Antibodies and reagents used for Western blotting analysis.

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Phosphoproteomics of CD2 signaling reveals AMPK-dependent regulation of lytic granule polarization in cytotoxic T cells

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AMPing up cytotoxicity

Cytotoxic T cells (CTLs) form an immune synapse at the interface formed with their target cells and then reorient intracellular lytic granules toward the contact site to ensure efficient killing. Zurli *et al.* found that stimulation of the receptor CD2 on CTLs by CD58 on target B cells enhanced T cell receptor signaling during immune synapse formation. Phosphoproteomics analysis showed that CD2 stimulation led to activation of the metabolism-regulating kinase AMPK on lysosomes, which promoted lytic granule translocation to the immune synapse. Together, these findings suggest that targeting the CD2-AMPK axis may enhance CTL activity.

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